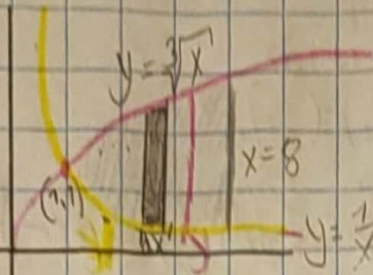


Tarea sección 6.1

1)



$$A = b \cdot h$$

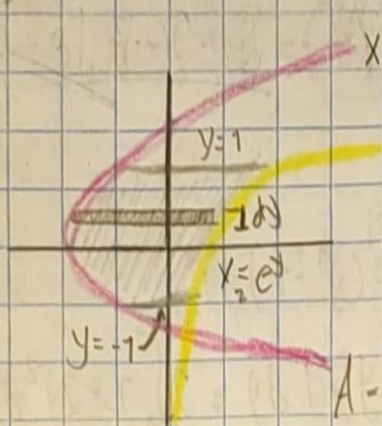
$$A = (y_2 - y_1) dx$$

$$A = \int_1^8 \left(\sqrt{x} - \frac{1}{x} \right) dx$$

$$A = \int_1^8 \sqrt{x} - \frac{1}{x} dx = \int_1^8 x^{1/2} dx - \int_1^8 \frac{1}{x} dx$$

$$A = \left[\frac{2}{3} x^{3/2} - \ln|x| \right]_1^8 = \left(\frac{2}{3} (8)^{3/2} - \ln(8) \right) - \left(\frac{2}{3} (1)^{3/2} - \ln(1) \right) = \frac{45}{4} - \ln(8)$$

3)



$$x = y^2 - 2$$

$$A = b \cdot h$$

$$A = (x_2 - x_1) dy$$

$$A = \int_{-1}^1 (e^y - (y^2 - 2)) dy$$

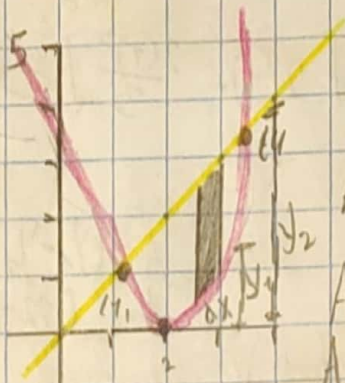
$$A = \int_{-1}^1 e^y - y^2 + 2 dy = \left[e^y - \frac{y^3}{3} + 2y \right]_{-1}^1$$

$$A = \left(e^1 - \frac{1^3}{3} + 2(1) \right) - \left(e^{-1} - \frac{(-1)^3}{3} + 2(-1) \right) = \left(e + \frac{5}{3} \right) - \left(\frac{1}{e} - \frac{5}{3} \right)$$

$$A = e - \frac{1}{e} - \frac{10}{3}$$

7) $y = (x-2)^2$, $y = x$

$$\begin{aligned} (x-2)^2 &= 0 \\ x-2 &= 0 \\ x &= 2 \\ y &= 0 \end{aligned}$$



$$A = b \cdot h$$

$$A = (y_2 - y_1) dx$$

$$A = \int_1^4 (x - (x^2 - 4x + 4)) dx$$

$$A = \int_1^4 x - x^2 + 4x - 4 dx = \left[\frac{5x^2}{2} - \frac{x^3}{3} - 4x \right]_1^4$$

$$A = \left(\frac{5}{2} (4)^2 - \frac{4^3}{3} - 4(4) \right) - \left(\frac{5}{2} (1)^2 - \frac{1^3}{3} - 4(1) \right) = \frac{8}{3} - \left(-\frac{11}{6} \right)$$

$$A = \frac{9}{2}$$

$$(x-2)^2 = x$$

$$x^2 - 4x + 4 = x$$

$$x^2 - 4x + 4 - x = 0$$

$$x^2 - 5x + 4 = 0$$

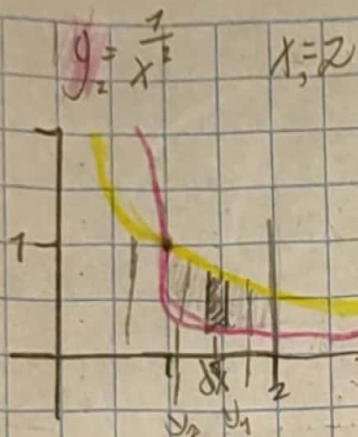
$$x(x-4) - 4(x-1) = 0$$

$$x = 1$$

$$x = 4$$

9) $y_1 = \frac{1}{x}$

$\frac{1}{x} = 0$
 $x = 0$



$A = \int (y_1 - y_2) dx$
 $A = \int_1^2 \left(\frac{1}{x} - \frac{1}{2} \right) dx$
 $A = \int_1^2 \left(\frac{1}{x} - \frac{1}{2} \right) dx = \int_1^2 \frac{1}{x} dx - \int_1^2 \frac{1}{2} dx$
 $A = \ln|x| - \frac{1}{2}x \Big|_1^2 = \ln 2 - \frac{1}{2}(2) - \left(\ln 1 - \frac{1}{2}(1) \right)$

$A = (\ln(2) - \frac{1}{2}) - (\ln(1) - \frac{1}{2}) = (\ln 2 - \frac{1}{2}) - (0 - \frac{1}{2}) = \ln(2) - \frac{1}{2}$

11) $x_1 = 1 - y^2$

$x_2 = y^2 - 1$

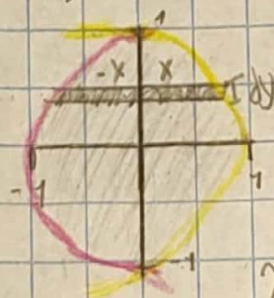
$A = \int (x_1 - x_2) dy$

$A = \int_{-1}^1 (1 - y^2) - (y^2 - 1) dy$

$A = 2 \int_0^1 (1 - y^2 - y^2 + 1) dy = 2 \int_0^1 (2 - 2y^2) dy$

$2 \left[2y - \frac{2y^3}{3} \right] \Big|_0^1$

$A = 2 \left[2(1) - \frac{2(1)^3}{3} \right] = \frac{8}{3}$



Intersecciones

$1 - y^2 = y^2 - 1$

$1 + 1 = y^2 + y^2$

$2 = 2y^2$

$\frac{2}{2} = y^2$

$\sqrt{1} = y$

$y = 1, x = 0$

$y = -1, x = 0$

13) $y_1 = 12 - x^2$

$y_2 = x^2 - 6$

$A = \int (y_1 - y_2) dx$

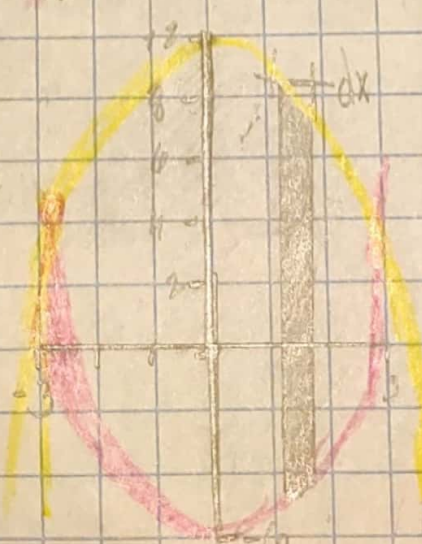
$A = \int_{-3}^3 (12 - x^2) - (x^2 - 6) dx$

$A = \int_{-3}^3 12 - x^2 - x^2 + 6 dx = \int_{-3}^3 18 - 2x^2 dx$

$A = 18x - \frac{2x^3}{3} \Big|_{-3}^3$

$A = \left(18(3) - \frac{2(3)^3}{3} \right) - \left(18(-3) - \frac{2(-3)^3}{3} \right)$

$A = 36 - (-36) = 72$



Intersecciones

$12 - x^2 = x^2 - 6$

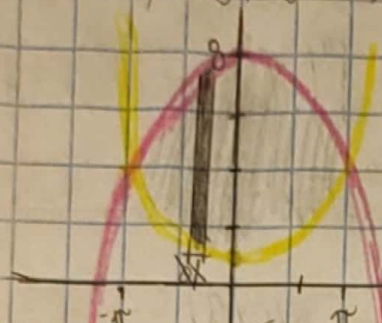
$12 + 6 = x^2 + x^2$

$18 = 2x^2$

$\sqrt{9} = x$

$\pm 3 = x$

15) $y_1 = \sec^2 x$, $y_2 = 8 \cos x$, $-\frac{\pi}{3} \leq x \leq \frac{\pi}{3}$



$$A = \int_{-\pi/3}^{\pi/3} (y_2 - y_1) dx$$

$$A = \int_{-\pi/3}^{\pi/3} (8 \cos x - \sec^2 x) dx$$

$$A = 8 \int_{-\pi/3}^{\pi/3} \cos x dx - \int_{-\pi/3}^{\pi/3} \sec^2 x dx$$

$$A = 8 \sin x - \tan x \Big|_{-\pi/3}^{\pi/3} = (8 \sin(\frac{\pi}{3}) - \tan(\frac{\pi}{3})) - (8 \sin(-\frac{\pi}{3}) - \tan(-\frac{\pi}{3}))$$

$$A = 3\sqrt{3} - (-3\sqrt{3}) = 6\sqrt{3}$$

17) $x_1 = 2y^2$, $x_2 = 4 + y^2$

distancia

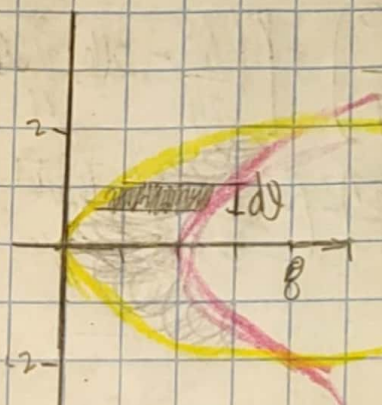
$$2y^2 = 4 + y^2$$

$$2y^2 - y^2 = 4$$

$$y^2 = 4$$

$$y = -2, x = 8$$

$$y = 2, x = 8$$



$$A = \int_{-2}^2 (x_2 - x_1) dy$$

$$A = 2 \int_{-2}^2 (4 + y^2 - 2y^2) dy$$

$$A = 2 \int_{-2}^2 (4 - y^2) dy$$

$$A = 2 \left(4y - \frac{y^3}{3} \right) \Big|_{-2}^2$$

$$A = 2 \left(4(2) - \frac{2^3}{3} \right) = \frac{32}{3}$$

19) $y_1 = \cos \pi x$, $y_2 = 4x^2 - 1$

$$A = \int_{-0.5}^{0.5} (y_1 - y_2) dx$$

$$A = \int_{-0.5}^{0.5} (\cos \pi x - (4x^2 - 1)) dx$$

$$A = \int_{-0.5}^{0.5} \cos \pi x dx - \int_{-0.5}^{0.5} (4x^2 - 1) dx$$

$$A = \int_{-0.5}^{0.5} \cos m \frac{dm}{\pi} - \left(\frac{4x^3}{3} - x \right) \Big|_{-0.5}^{0.5}$$

$$A = \frac{1}{\pi} \sin(\pi x) - \left(\frac{4x^3}{3} - x \right) \Big|_{-0.5}^{0.5}$$

$$A = \left(\frac{1}{\pi} \sin(\pi \cdot \frac{1}{2}) - \frac{4(\frac{1}{2})^3}{3} + \frac{1}{2} \right) - \left(\frac{1}{\pi} \sin(\pi \cdot -\frac{1}{2}) - \frac{4(-\frac{1}{2})^3}{3} - \frac{1}{2} \right)$$

$$A = \left(\frac{1}{\pi} - \frac{1}{6} + \frac{1}{2} \right) - \left(-\frac{1}{\pi} + \frac{1}{6} - \frac{1}{2} \right) = \frac{2}{\pi} - \frac{2}{3} + \frac{2}{3} = \frac{2}{\pi}$$

$$m = \pi x$$

$$dm = \pi dx$$

$$\frac{dm}{\pi} = dx$$

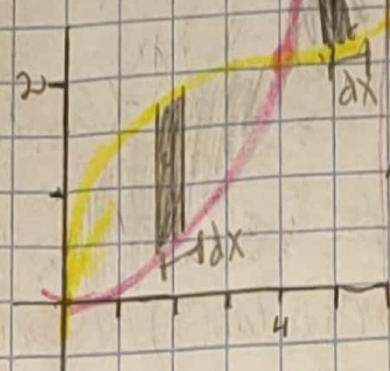
~~A~~

23) $y_1 = \sqrt[3]{2x}$, $y_2 = \frac{1}{8}x^2$ $0 \leq x \leq 6$

intersection

$\sqrt[3]{2x} = \frac{1}{8}x^2$
 $8\sqrt[3]{2x} = x^2$
 $512 \cdot 2x = x^6$
 $1024x = x^6$

$1024x - x^6 = 0$
 $x(1024 - x^5) = 0$
 $x = 0$
 $x = 4$



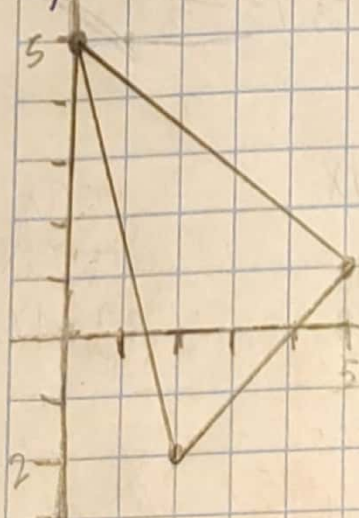
$A = \int_0^4 (\sqrt[3]{2x} - \frac{1}{8}x^2) dx + \int_4^6 (\frac{1}{8}x^2 - \sqrt[3]{2x}) dx$
 $A = \int_0^4 \sqrt[3]{2x} - \frac{1}{8}x^2 dx$

$A = \frac{3x\sqrt[3]{2x}}{4} - \frac{x^3}{24} \Big|_0^4 + \frac{x^3}{24} - \frac{3x\sqrt[3]{2x}}{4} \Big|_4^6$

$A = \left(\frac{3(4)\sqrt[3]{2(4)}}{4} - \frac{4^3}{24} \right) + \left[\left(\frac{6^3}{24} - \frac{3(6)\sqrt[3]{2(6)}}{4} \right) - \left(\frac{4^3}{24} - \frac{3(4)\sqrt[3]{2(4)}}{4} \right) \right]$

$A = (6\sqrt[3]{8} - \frac{8}{3}) + \left[(9 - \frac{9\sqrt[3]{12}}{2}) - (-\frac{10}{3}) \right] = 6\sqrt[3]{8} - \frac{8}{3} + (\frac{27}{3} - \frac{9\sqrt[3]{12}}{2})$

33) (0, 5), (2, -2), (5, 1)



$m = \frac{-2-5}{2-0} = -\frac{7}{2}$ $y - 0 = -\frac{7}{2}(x - 3)$
 $y = -\frac{7}{2}x + \frac{21}{2}$

$m = \frac{1-2}{5-2} = -\frac{1}{3}$ $y - 2 = -\frac{1}{3}(x - 2)$
 $y = -\frac{1}{3}x + \frac{8}{3}$

$m = \frac{1-5}{5-0} = -\frac{4}{5}$ $y - 5 = -\frac{4}{5}(x - 0)$
 $y = -\frac{4}{5}x + 5$

$A = \int_0^2 (-\frac{4}{5}x + 5) - (-\frac{7}{2}x + \frac{21}{2}) dx + \int_2^5 (-\frac{4}{5}x + 5) - (x - 4) dx$

$= \left[-\frac{2}{5}x^2 + 5x + \frac{7}{4}x^2 - \frac{21}{2}x \right]_0^2 + \left[-\frac{1}{2}x^2 + 5x - \frac{x^2}{2} + 4x \right]_2^5$

$$\text{* DUKI}$$

$$\left(\frac{-2(5)^2}{5} + 5(5)\right) - \left(\frac{5^2}{2} + 4(5)\right) - \left(\frac{-2(2)^2}{5} - 5(2) - \frac{5(2)^2}{2} + 1(2)\right) = \frac{8^2+27}{10} - \frac{27}{5} - \frac{27}{2}$$

$$35) \int_0^4 \sqrt{x+2} - x \, dx$$

$$\int_0^2 \sqrt{x+2} - x + \int_2^4 -\sqrt{x+2} = \frac{16-4\sqrt{2}-2}{3} + \left[-4\sqrt{2} + \frac{24}{3}\right] =$$

$$\frac{14}{3} - \frac{4\sqrt{2}}{3} - 2 - 4\sqrt{2} + \frac{24}{3} = \frac{44}{3} - \frac{4\sqrt{2}}{3} - 4\sqrt{2}$$

$$37) y=6, \quad y=x^2, \quad x=4$$

$$x^2=4 = x\sqrt{4} \quad A = \int_{-2}^2 4x^2 \, dx = 4x - \frac{x^3}{3} \Big|_{-2}^2$$

$$\left[4(2) - \frac{2^3}{3}\right] - \left[4(-2) - \frac{(-2)^3}{3}\right] = 8 - \frac{8}{3} + 8 - \frac{8}{3} = \frac{32}{3}$$

$$\frac{32}{3} \cdot \frac{1}{2} - \frac{16}{3} x^2 = 6 \quad x = \pm\sqrt{5}, -0.6$$

$$\int_{-b}^b (b-x^2) \, dx = \frac{16}{3} = 6x - \frac{x^3}{3} = \left(6\sqrt{6} - \frac{(\sqrt{6})^3}{3}\right) - \left(6(-\sqrt{6}) - \frac{(-\sqrt{6})^3}{3}\right)$$

$$= 6\sqrt{6} - \frac{\sqrt{6}^3}{3} - \left(-6\sqrt{6} - \frac{\sqrt{6}^3}{3}\right) = \frac{16}{3}$$

$$= \sqrt{6}^3 \cdot \frac{\sqrt{6}^3}{3} + \sqrt{6}^3 \cdot \frac{\sqrt{6}^3}{3} = \frac{16}{3} = \frac{4\sqrt{6}^3}{3} = \frac{16}{3} \cdot \frac{3}{4} = 4 = \sqrt{6}^2$$

$$b^3 = 9^2 = 6 = \sqrt[3]{16}$$

$$59) y=x^2-c^2 \quad y=c^2-x^2 \quad \text{u 576}$$

$$x^2-c^2 = c^2-x^2$$

$$x^2 + x^2 = c^2 + c^2$$

$$2x^2 = 2c^2$$

$$x^2 = c^2$$

$$x = \pm\sqrt{c^2}$$

$$\int_{-c}^c c^2 - x^2 - (x^2 - c^2) \, dx = 576$$

$$\int_{-c}^c (c^2 - x^2 - x^2 + c^2) \, dx = 576$$

$$\left(\frac{2c^3}{3} - 2x^2\right) \Big|_{-c}^c = 576 \Big|_{-c}^c$$

$$2c^2x - \frac{2x^3}{3} \Big|_{-c}^c = \left(2c^2(c) - \frac{2c^3}{3}\right) - \left(2c^2(-c) - 2\frac{(-c)^3}{3}\right) = 576$$

$$\left(2c^3 - \frac{2c^3}{3}\right) - \left(-2c^3 + \frac{2c^3}{3}\right) = 576$$

$$2c^3 - \frac{2c^3}{3} + 2c^3 - \frac{2c^3}{3} = 576$$

$$\frac{8c^3}{3} = 576$$

$$c^3 = \frac{3}{8}(576) = 216$$

$$c = \sqrt[3]{216} = c = \pm 6$$

(6) Data